

Infinite Series

Complex Analysis

Morales Carmona Ronny

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Infinite Series

Tests for
Convergence

Cauchy Criterion

Cauchy Products

Comparison Tests

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Convergent SeriesRearrangements of
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Definition

If $Z = (z_n)$ is a sequence in $\mathbb{C}$, then the **formal infinite series** (or simply the **series**) **generated by Z** is the sequence $S := (s_k)$ defined by

$$s_1(z) := z_1$$

$$s_2(z) := s_1(z) + z_2 \quad (= z_1 + z_2)$$

$$\vdots$$

$$s_k(z) := s_{k-1}(z) + z_k \quad (= z_1 + z_2 + \cdots + z_k)$$

$$\vdots$$

The numbers z_n are called the **terms** of the series and the numbers s_k are called the **partial sums** of this series. If $\lim S$ exists, we say that this series is **convergent** and call this limit the **sum** or the **value** of this series. If this limit does not exist, we say that the series S is **divergent**. It is convenient to use symbols such as

$$\sum (z_n) \quad \text{or} \quad \sum z_n \quad \text{or} \quad \sum_{n=1}^{\infty} z_n$$

to denote both the formal infinite series S generated by the sequence Z (z_n) and also to denote the value $\lim S$, in case this limit exists.

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Example 1.1

Consider the sequence (z^n) where $z \in \mathbb{C}$, which generates the **geometric series**:

$$\sum_{n=0}^{\infty} z^n = 1 + z + z^2 + \cdots + z^n + \cdots .$$

Show that if $|z| < 1$, the series converges to

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1 - z}.$$

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Example 1.2

- a) Consider the series generated by the sequence (n) ; that is, the series

$$\sum_{n=0}^{\infty} n = 1 + 2 + 3 + \cdots .$$

Show that the series is divergent.

- b) Consider the series generated by (x_n) with $x_n = (-1)^n$; that is, the series

$$\sum_{n=0}^{\infty} (-1)^n = (+1) + (-1) + (+1) + (-1) + \cdots .$$

Determine whether the series is convergent or divergent.

- c) Consider the series

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} .$$

Show that this series converges to 1.

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Theorem 1.3 (Properties of Series)

- i) If $\sum a_n$ converges to A , and $\sum b_n$ converges to B , then $\sum(a_n + b_n)$ is also a convergent series, and converges to $A + B$.
- ii) If $\sum a_n$ converges to A , and c is a real number, then $\sum ca_n$ is also a convergent series, and converges to cA .
- iii) Let $k \in \mathbb{Z}^+$. If one of the two series $\sum_0^\infty a_n$ or $\sum_k^\infty a_n$ are convergent, then the other one is also, and we have the identity

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{k-1} a_n + \sum_{n=k}^{\infty} a_n.$$

- iv) Let $\sum a_n$ converge to A , and let k be an integer. Then

$$\sum_{n=k}^{\infty} a_{n-k} = A.$$

Theorem 1.4 (The n th Term Test)

If the series $\sum z_n$ converges, then $\lim(z_n) = 0$.

Theorem 1.5 (Leibniz's Criterion for Alternating Series)

Let (x_n) be a sequence of real numbers which are nonnegative and decreasing. Then the series $\sum (-1)^n (x_n)$ is convergent if and only if $\lim(x_n) = 0$.

Theorem 1.6 (Cauchy Criterion for Series)

The series $\sum z_n$ converges if and only if for every $\varepsilon > 0$ there exists $M(\varepsilon) \in \mathbb{N}$ such that if $m > n \geq M(\varepsilon)$, then

$$|s_m - s_n| = |z_{n+1} + z_{n+2} + \cdots + z_m| < \varepsilon.$$

Theorem 1.7

Let (x_n) be a sequence of nonnegative real numbers. Then the series $\sum x_n$ converges if and only if the sequence $S = (s_k)$ of partial sums is bounded. In this case,

$$\sum_{n=1}^{\infty} x_n = \lim(s_k) = \sup\{s_k : k \in \mathbb{N}\}.$$

Example 1.8

The **harmonic series** (which receives its musical name from the fact that the wavelengths of the overtones of a vibrating string are $1/2, 1/3, 1/4, \dots$, of the string's fundamental length), given by

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

is divergent.

Example 1.9

The p -series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges for $p > 1$ and diverges for $0 < p \leq 1$.

Example 1.10

The **alternating harmonic series**, given by

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

is convergent.

Definition

Let $\sum a_n$ and $\sum b_n$ be two complex series. Their **Cauchy product** is defined to be the series

$$\sum_{n=0}^{\infty} c_n \quad \text{with} \quad c_n = \sum_{k=0}^n a_k b_{n-k}.$$

Theorem 1.11 (Merten's Theorem)

Let $\sum a_n$ and $\sum b_n$ be two convergent series such that one of them converges absolutely. Then their Cauchy product converges to the product $\sum a_n \sum b_n$.

Theorem 1.12 (Comparison Test)

Let (x_n) and (y_n) be real sequences and suppose that for some $K \in \mathbb{N}$ we have

$$0 \leq x_n \leq y_n \quad \text{for } n \geq K.$$

- i) Then the convergence of $\sum y_n$ implies the convergence of $\sum x_n$.
- ii) The divergence of $\sum x_n$ implies the divergence of $\sum y_n$.

Theorem 1.13 (Limit Comparison Test)

Suppose that (x_n) and (y_n) are strictly positive sequences and suppose that the following limit exists in $\mathbb{R}$:

$$r := \lim \left(\frac{x_n}{y_n} \right).$$

- i) *If $r \neq 0$ then $\sum x_n$ is convergent if and only if $\sum y_n$ is convergent.*
- ii) *If $r = 0$ and if $\sum y_n$ is convergent, then $\sum x_n$ is convergent.*

Definition

Let $Z = (z_n)$ be a complex sequence. We say that the series $\sum z_n$ is **absolutely convergent** if the series $\sum |z_n|$ is convergent in $\mathbb{C}$. A series is said to be **conditionally (or nonabsolutely) convergent** if it is convergent, but it is not absolutely convergent.

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Theorem 1.14

If a series in $\mathbb{C}$ is absolutely convergent, then it is convergent.

Definition

A series $\sum w_k$ in $\mathbb{C}$ is a **rearrangement** of a series $\sum z_n$ if there is a bijection π of $\mathbb{N}$ onto $\mathbb{N}$ such that $w_k = z_{\pi(k)}$ for all $k \in \mathbb{N}$.

Theorem 1.15 (Riemann Rearrangement Theorem)

Let $\sum x_n$ be a conditionally convergent real series and $L \in \mathbb{R}$ be any real number. Then, there is a rearrangement $\sum y_n$ of the series $\sum x_n$ such that it converges to L , namely, $\sum x_{\pi(n)} = L$ for some bijection $\pi : \mathbb{N} \rightarrow \mathbb{N}$.

Theorem 1.16 (Limit Comparison Test II)

Suppose that $X = (x_n)$ and $Y = (y_n)$ are nonzero real sequences and suppose that the following limit exists in $\mathbb{R}$:

$$r := \lim \left| \frac{x_n}{y_n} \right|.$$

- i) *If $r \neq 0$, then $\sum x_n$ is absolutely convergent if and only if $\sum y_n$ is absolutely convergent.*
- ii) *If $r = 0$ and if $\sum y_n$ is absolutely convergent, then $\sum x_n$ is absolutely convergent.*

Theorem 1.17 (Cauchy's n th Root Test)

Let (z_n) be a sequence in $\mathbb{C}$.

i) If there exist $\mathbb{R} \ni r < 1$ and $K \in \mathbb{N}$ such that

$$|z_n|^{1/n} \leq r \quad \text{for } n \geq K,$$

then the series $\sum z_n$ is absolutely convergent.

ii) If there exists $K \in \mathbb{N}$ such that

$$|z_n| \geq 1 \quad \text{for } n \geq K,$$

then the series is divergent.

Corollary 1.18

Let (z_n) be a sequence in $\mathbb{C}$ and suppose that the limit

$$r := \lim |z_n|^{1/n}$$

exists in $\mathbb{R}$. Then $\sum z_n$ is absolutely convergent when $r < 1$ and is divergent when $r > 1$.

Theorem 1.19 (D'Alembert's Ratio Test)

Let (z_n) be a sequence of nonzero complex numbers.

i) If there exist $0 < r < 1$ and $K \in \mathbb{N}$ such that

$$\left| \frac{z_{n+1}}{z_n} \right| \leq r \quad \text{for } n \geq K,$$

then the series $\sum z_n$ is absolutely convergent.

ii) If there exists $K \in \mathbb{N}$ such that

$$\left| \frac{z_{n+1}}{z_n} \right| \geq 1 \quad \text{for } n \geq K,$$

then the series $\sum z_n$ is divergent.

Corollary 1.20

Let (z_n) be a nonzero sequence in $\mathbb{C}$ and suppose that the limit

$$r := \lim \left| \frac{z_{n+1}}{z_n} \right|$$

exists in $\mathbb{R}$. Then $\sum z_n$ is absolutely convergent when $r < 1$ and is divergent when $r > 1$.

Definition

If $Z = (z_1, z_2, \dots, z_n, \dots)$ is a sequence of complex numbers and if m is a given natural number, then the **m -tail** of Z is the sequence

$$Z_m := (z_{m+n} : n \in \mathbb{N}).$$

Theorem 1.21

Let $Z := (z_n : n \in \mathbb{N})$ be a sequence of complex numbers and let $m \in \mathbb{N}$. Then the m -tail of $Z_m := (z_{m+n} : n \in \mathbb{N})$ of Z converges if and only if Z converges. In this case, $\lim Z_m = \lim Z$.

Definition

Let S be a nonempty subset of $\mathbb{R}$.

- i) The set S is said to be **bounded above** if there exists a number $u \in \mathbb{R}$ such that $s \leq u$ for all $s \in S$. Each number u is called an **upper bound** of S .
- ii) The set S is said to be **bounded below** if there exists a number $w \in \mathbb{R}$ such that $w \leq s$ for all $s \in S$. Each number w is called a **lower bound** of S .
- iii) A set is said to be **bounded** if it is both bounded above and bounded below. A set is said to be **unbounded** if it is not bounded.

Definition

Let S be a nonempty subset of $\mathbb{R}$.

- i) If S is bounded above, then a number u is said to be a **supremum** (or a **least upper bound**) of S if it satisfies the conditions:
 - a) u is an upper bound of S , and
 - b) if v is any upper bound of S , then $u \leq v$.
- ii) If S is bounded below, then a number w is said to be an **infimum** (or a **greatest lower bound**) of S if it satisfies the conditions:
 - a') w is a lower bound of S , and
 - b') if t is any lower bound of S , then $t \leq w$.

If the supremum or the infimum of a set S exists, we will denote them by

$$\sup S \quad \text{and} \quad \inf S.$$

Definition

Let $X = (x_n)$ be a bounded sequence in $\mathbb{R}$ and let X_m be the m -tail of X . We define U to be the sequence

$$U := (u_m), \quad u_m := \sup X_m,$$

similarly, we define L to be the sequence

$$L := (\ell_m), \quad \ell_m := \inf X_m.$$

The **limit superior** of X is defined as the limit $\lim U$ of the sequence U . It is denoted by

$$\limsup(x_n) \quad \text{or} \quad \limsup X \quad \text{or} \quad \overline{\lim}(x_n).$$

The **limit inferior** of X is the limit $\lim L$ of the sequence L . It is denoted by

$$\liminf(x_n) \quad \text{or} \quad \liminf X \quad \text{or} \quad \underline{\lim}(x_n).$$

Remark

If a sequence $X = (x_n)$ is not bounded, then we define

$$\limsup X = \infty$$

if it is not bounded from above, and

$$\liminf X = -\infty$$

if it is not bounded from below.

Theorem 1.22

Let (x_n) be a bounded sequence in $\mathbb{R}$. Then (x_n) is convergent if and only if $\limsup x_n = x = \liminf x_n$. In this case, $x = \lim x_n$.

Lemma 1.23

Let (a_n) be a sequence in $\mathbb{R}$ and let

$$r := \limsup a_n.$$

Then for any $\rho > r$ there exists $N \in \mathbb{N}$ such that $a_n < \rho$ for all $n \geq N$.

Theorem 1.24 (Cauchy's n th root test (Strong Version))

Let (z_n) be a sequence of complex numbers and let

$$r := \limsup |z_n|^{1/n}.$$

Then $\sum z_n$ is absolutely convergent if $r < 1$, and diverges if $r > 1$.

Theorem 1.25 (D'Alembert's Ratio Test (Strong version))

Let (z_n) be a sequence of nonzero complex numbers and let

$$r := \liminf \left| \frac{z_{n+1}}{z_n} \right|, \quad R := \limsup \left| \frac{z_{n+1}}{z_n} \right|.$$

Then $\sum z_n$ converges absolutely if $R < 1$, and diverges if $r > 1$.